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FDE Cycle & Systems Software

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OCR GCSE Computer Science — **Systems Software (J277/01 Section 1.5)**. This topic covers the purpose and functions of operating systems, how the OS manages hardware resources and coordinates the CPU, and the role of utility software in maintaining system performance and security.

Exam questions typically ask you to **describe** functions of an operating system, **explain** how memory management or multitasking works, and **identify** appropriate utility software for given scenarios. Questions range from 1-mark definitions to 4-mark explanations requiring you to link OS functions to real-world situations.

What you must know

This checklist covers the key exam-visible concepts. You should be able to explain each point and give an example where appropriate.

- **State** the purpose of an operating system and why it is needed.
- **Describe** the key functions of an operating system: user interface, memory management, multitasking, peripheral management, user management, file management.
- **Explain** how the operating system manages memory, including the allocation of memory to applications and the use of virtual memory.
- **Explain** how multitasking allows multiple programs to run simultaneously.

- **Describe** the purpose of utility software and give examples: encryption, defragmentation, data compression, backup.
- **Explain** why utility software is needed to perform tasks that may not be carried out by an operating system.

> **Specification note.** OCR J277 Section 1.5 explicitly requires understanding of operating systems and their functions, plus utility software. The September 2024 specification update clarified that utility software performs additional tasks that may not be carried out by an operating system, and added memory allocation to applications as explicit content.

Toolkit

Operating system — definition

An **operating system (OS)** is software that manages computer hardware and provides services for application programs. It acts as an **interface between the user and the hardware**, hiding the complexity of the hardware and allowing users and programs to interact with the computer without needing to know how the hardware works.

The six key OS functions

Function	What the OS does	Example
User interface	Provides a way for users to interact with the computer (GUI, command line, menu, etc.)	Windows desktop with icons, Start menu, and taskbar
Memory management	Allocates RAM to applications, moves data between RAM and virtual memory, prevents programs from accessing each other's memory	When you open Chrome, the OS allocates RAM for it
Multitasking	Allows multiple programs to run at the same time by rapidly switching CPU time between them	Listening to music while writing a document
Peripheral management	Controls input/output devices using device drivers; handles communication between hardware and software	Installing a printer driver so the OS can send print jobs
User management	Manages user accounts, passwords, permissions, and access levels	Admin accounts can install software; standard users cannot
File management	Organises files into folders, controls file naming, handles read/write operations, manages file permissions	Creating folders, renaming files,

Function	What the OS does	Example
		setting files as read-only

Utility software — the four key types

Utility	Purpose	When/why it is used
Encryption software	Converts data into an unreadable format using a key; only those with the key can decrypt it	Protecting sensitive data; securing files before sending over a network
Defragmentation software	Reorganises fragmented files on a hard disk so each file is stored in contiguous sectors	Improving read/write speed on HDDs; not needed for SSDs
Data compression software	Reduces file size by removing redundancy (lossless) or non-essential data (lossy)	Saving storage space; faster file transfer over networks
Backup software	Creates copies of files/data that can be restored if the original is lost, corrupted, or deleted	Disaster recovery; protecting against ransomware, hardware failure, or accidental deletion

Memory management and virtual memory

The OS is responsible for **allocating memory to applications**. When a program starts, the OS assigns it a portion of RAM. The OS ensures programs cannot access each other's memory space, preventing crashes and security issues.

When RAM is full, the OS uses **virtual memory** — it moves inactive data from RAM to a section of secondary storage (the swap file or page file). This allows more programs to run than would fit in physical RAM alone, but accessing virtual memory is slower than accessing RAM.

Exam tip: When asked about OS functions, always name the specific function (e.g. "memory management") and then describe what the OS actually does. Vague answers like "controls the computer" will not score marks.

1. Operating systems

An **operating system (OS)** is the most important piece of software on a computer. It manages all the hardware and provides a platform for application software to run. Without an operating system, you would need to write complex code just to display text on screen or read from a keyboard.

1.1 Purpose of the operating system

The operating system has several key purposes:

Hardware abstraction: The OS hides the complexity of the hardware from users and application programs. You do not need to know how to send electrical signals to a hard drive — you just click "Save" and the OS handles everything.

Resource management: The OS manages all the computer's resources — CPU time, memory, storage, and peripherals — ensuring they are shared fairly between programs and users.

Providing a platform: Application software (like word processors, games, and browsers) runs on top of the operating system. The OS provides services that applications can use, such as file access, network connections, and graphics display.

User interface: The OS provides a way for users to interact with the computer. This might be a graphical user interface (GUI) with windows, icons, and menus, or a command-line interface (CLI) where users type commands.

1.2 Types of operating system

Different devices use different types of operating system:

- **Desktop/laptop OS:** Windows, macOS, Linux — designed for general-purpose computing with full GUI
- **Mobile OS:** Android, iOS — optimised for touchscreen devices with limited power
- **Server OS:** Windows Server, Linux distributions — designed to handle many users and network services
- **Embedded OS:** Real-time operating systems in devices like washing machines, car systems — minimal, dedicated to specific tasks

1.3 The OS and the CPU

The operating system works closely with the CPU. When you run multiple programs, the OS decides which program gets to use the CPU at any moment. It schedules processes, switches between them rapidly (context switching), and ensures each program gets a fair share of processing time.

When the CPU executes the fetch-decode-execute cycle, it is running instructions from either the OS itself or from an application. The OS controls when each application's instructions are fetched and executed.

1.4 The OS as an interface

Think of the OS as sitting between the user/applications and the hardware:

- **Above the OS:** Users and application software
- **The OS:** Translates requests into hardware operations
- **Below the OS:** Hardware (CPU, RAM, storage, peripherals)

When an application wants to save a file, it asks the OS. The OS then works out where on the disk to store the data, sends the correct signals to the storage device, and reports back to the application when the operation is complete.

Exam tip: A common question asks "Why is an operating system needed?" Strong answers mention: managing hardware, providing a user interface, allowing multiple programs to run, and providing services for applications.

Worked Example 1 — Purpose of an OS

Question: State two reasons why a computer needs an operating system. [2 marks]

Model answer:
1. To manage the hardware/resources of the computer **(1)**.
2. To provide an interface between the user and the hardware **(1)**.

Exam tip: Other acceptable answers include: to allow multiple programs to run, to manage memory, to provide a platform for applications. Give two distinct reasons.

Worked Example 2 — OS as an interface

Question: Explain how the operating system acts as an interface between the user and the hardware. [2 marks]

Model answer:
The OS translates user actions (like clicking an icon) into hardware operations **(1)**. This means the user does not need to know how the hardware works to use the computer **(1)**.

Exam tip: The key idea is "hiding complexity" — the OS handles the technical details so users and applications don't have to.

Worked Example 3 — Identifying OS types

Question: Give one example of a mobile operating system and one example of a desktop operating system. [2 marks]

Model answer:
Mobile: Android / iOS **(1)**.
Desktop: Windows / macOS / Linux **(1)**.

Exam tip: Make sure you match the OS type correctly. Windows is a desktop OS, not a mobile OS.

Worked Example 4 — OS and applications

Question: Explain why application software needs an operating system to run. [2 marks]

Model answer:
The OS provides services that applications use, such as file access, memory allocation, and display output **(1)**. Without the OS, applications would have to communicate directly with hardware, which would be complex and impractical **(1)**.

Exam tip: Think of the OS as providing a "platform" — it handles the hard work so applications can focus on their specific tasks.

2. Functions of the operating system

The operating system performs many functions to manage the computer's resources and provide services. You need to know six key functions in detail.

2.1 User interface

The OS provides a **user interface** that allows people to interact with the computer. There are several types:

- **GUI (Graphical User Interface):** Uses windows, icons, menus, and pointers (WIMP). Easy to use but requires more processing power.
- **CLI (Command Line Interface):** Users type text commands. More efficient for experts but harder to learn.
- **Menu-driven interface:** Users select from a list of options. Common in ATMs and simple devices.
- **Natural language interface:** Users speak or type in natural language (e.g. voice assistants).

2.2 Memory management

The OS is responsible for **memory management** — controlling how RAM is used:
Allocation: When a program starts, the OS allocates a portion of RAM for it to use. The OS tracks which memory is in use and which is free.
Protection: The OS prevents programs from accessing memory allocated to other programs. This stops one program from crashing another or accessing sensitive data.
Virtual memory: When RAM is full, the OS moves inactive data to secondary storage (the swap file). This allows more programs to run than would fit in RAM, but virtual memory is slower to access.

2.3 Multitasking

Multitasking allows multiple programs to appear to run simultaneously. The OS achieves this by:

- Dividing CPU time into small slices
- Rapidly switching between programs (context switching)
- Giving each program a share of processing time

Because the switching happens so fast (thousands of times per second), it appears that all programs are running at the same time. In reality, a single-core CPU can only execute one instruction at a time.

2.4 Peripheral management

The OS manages **peripheral devices** (printers, keyboards, mice, etc.) using **device drivers**:

Device drivers: Small programs that translate OS commands into instructions the specific hardware understands. Each device needs its own driver.

Plug and Play: Modern operating systems can detect new devices and automatically install appropriate drivers.

Spooling: For devices like printers, the OS uses a queue (spool) to manage multiple print jobs, allowing users to continue working while printing happens in the background.

2.5 User management

The OS provides **user management** features:

- **User accounts:** Each user has their own username and password
- **Access levels:** Administrators have full control; standard users have limited permissions
- **File permissions:** Users can only access files they have permission to view
- **Profiles:** Each user has their own settings, desktop, and files

2.6 File management

The OS handles **file management**:

- **Organisation:** Files stored in hierarchical folder structures
- **Naming:** Rules for valid file names and extensions
- **Operations:** Create, delete, copy, move, rename files
- **Permissions:** Read-only, hidden, system file attributes
- **Allocation:** Tracking which sectors on the disk belong to which file

> **Exam focus:** Memory management and multitasking are frequently tested. Be prepared to explain how the OS allocates memory, what virtual memory is, and how the OS switches between running programs.

Worked Example 5 — Memory management

Question: Describe how the operating system manages memory when multiple programs are running. [3 marks]

Model answer:
The OS allocates a portion of RAM to each program when it starts **(1)**. The OS prevents each program from accessing memory belonging to other programs **(1)**. If RAM becomes full, the OS moves inactive data to virtual memory on secondary storage **(1)**.

Exam tip: For memory management questions, cover: allocation, protection, and virtual memory. These are the three key aspects.

Worked Example 6 — Multitasking

Question: Explain how an operating system allows multiple programs to run at the same time on a single-core processor. [3 marks]

Model answer:
The OS divides CPU time into small slices/time slots **(1)**. It rapidly switches between programs, giving each a share of processing time **(1)**. This happens so fast that it appears all programs are running simultaneously **(1)**.

Exam tip: The key phrase is "time slicing" or "context switching". Emphasise that programs take turns using the CPU, creating the illusion of simultaneous execution.

Worked Example 7 — Peripheral management

Question: Explain the role of device drivers in peripheral management. [2 marks]

Model answer:
Device drivers are programs that translate operating system commands into instructions the specific hardware device understands **(1)**. Each peripheral requires its own driver because different devices have different instruction sets **(1)**.

Exam tip: Think of drivers as "translators" between the OS and the hardware. The OS speaks one language; each device speaks its own.

Worked Example 8 — Virtual memory

Question: Explain what virtual memory is and why it is used. [3 marks]

Model answer:
Virtual memory is a section of secondary storage that is used as if it were RAM **(1)**. When RAM is full, the OS moves inactive data to virtual memory to free up space **(1)**. This allows more programs to run than would fit in physical RAM alone **(1)**.

> **Exam tip:** Always mention that virtual memory is slower than RAM. If asked for a drawback, this is the key point — programs run more slowly when data has to be swapped between RAM and virtual memory.

3. Utility software

Utility software consists of programs designed to help maintain, optimise, and secure a computer system. Utility software performs tasks that the operating system may not carry out, or provides enhanced versions of OS functions.

3.1 Encryption software

Encryption software converts data into an unreadable format (ciphertext) using a mathematical algorithm and a key. Only someone with the correct key can decrypt the data back to its original form.

Uses:

- Protecting sensitive files stored on a computer
- Securing data before sending over a network
- Encrypting entire drives to protect against theft
- Protecting passwords and financial information

If encrypted data is intercepted or stolen, it is useless without the decryption key.

3.2 Defragmentation software

Over time, files on a **hard disk drive (HDD)** become fragmented — stored in non-contiguous sectors scattered across the disk. This happens because files are saved in whatever free space is available.

Defragmentation software reorganises the disk so that each file is stored in contiguous sectors. This reduces the time the read/write head needs to move around the disk, improving performance.

Important: SSDs do not need defragmentation because they have no moving parts. In fact, defragmenting an SSD can reduce its lifespan by causing unnecessary writes.

3.3 Data compression software

Compression software reduces file sizes:

- **Lossless compression:** Reduces size without losing any data. The original file can be perfectly reconstructed. Used for documents, programs, and data files.
- **Lossy compression:** Reduces size by permanently removing some data. Used for images, audio, and video where small quality losses are acceptable.

Benefits: Saves storage space, faster file transfer, reduced bandwidth usage.

3.4 Backup software

Backup software creates copies of files that can be restored if the original is lost, corrupted, or deleted.

Types of backup:

- **Full backup:** Copies all selected files. Takes longest but easiest to restore.
- **Incremental backup:** Only copies files changed since the last backup. Faster but requires all previous backups to restore.
- **Differential backup:** Copies files changed since the last full backup. Balance of speed and simplicity.

Why backups are needed:

- Protection against hardware failure
- Recovery from accidental deletion
- Protection against ransomware attacks
- Disaster recovery (fire, flood, theft)

3.5 Why utility software is needed

The operating system provides basic functionality, but utility software offers:

- **Specialised functions:** Tasks like encryption or defragmentation may not be included in the OS
- **Enhanced features:** Third-party utilities may be more powerful than built-in OS tools
- **System maintenance:** Keeping the computer running efficiently over time
- **Security:** Protecting data beyond what the OS provides by default

> **Exam tip:** The 2024 specification update emphasises that "utility software is needed to perform additional tasks that may not be carried out by an operating system." Be prepared to explain why a specific utility would be used alongside the OS.

Worked Example 9 – Encryption

Question: A company stores customer data on laptops. Explain why encryption software should be used. [3 marks]

Model answer:
Encryption converts the data into an unreadable format **(1)**. If a laptop is lost or stolen, the data cannot be accessed without the decryption key **(1)**. This protects customer data and helps the company comply with data protection laws **(1)**.

> **Exam tip:** Always link encryption to a scenario. The key benefit is that stolen/intercepted data is useless without the key.

Worked Example 10 – Defragmentation

Question: Explain what defragmentation software does and why it improves the performance of a hard disk drive. [3 marks]

Model answer:
Defragmentation rearranges files so they are stored in contiguous/adjacent sectors **(1)**. On a fragmented disk, the read/write head has to move to different locations to read parts of the same file **(1)**. After defragmentation, the head moves less, so files are read/written faster **(1)**.

Exam tip: If asked about SSDs, state that defragmentation is not needed because SSDs have no moving parts and can access any location equally fast.

Worked Example 11 — Backup and compression scenario

Question: A school backs up student work to cloud storage each night. Explain why the school should use both backup software and compression software. [4 marks]

Model answer:
Backup software: Creates copies of student work that can be restored if the original files are lost, corrupted, or accidentally deleted **(1)**. Protects against data loss from hardware failure, ransomware, or other disasters **(1)**.
Compression software: Reduces the size of the backup files **(1)**. This means the backup takes less time to upload, uses less cloud storage space, and reduces bandwidth/costs **(1)**.

Exam tip: When a question mentions two utilities, give separate points for each. Explain both the purpose and a specific benefit for the scenario given.

Worked Example 12 — Extended analysis (6-mark style)

Question: A small business is setting up new computers. Discuss the importance of the operating system's memory management and multitasking functions for the business, and explain why the business might also need utility software. [6 marks]

Plan:
- Explain memory management and its importance
- Explain multitasking and its importance
- Explain why utility software is also needed

Model answer:
Memory management is important because business users typically run multiple applications simultaneously — email, spreadsheets, databases, and browsers. The OS must allocate RAM to each application efficiently. If RAM becomes full, the OS uses virtual memory to ensure all applications can continue running, although this may slow performance.
Multitasking allows employees to work productively by running several programs at once. The OS rapidly switches between applications, giving each a share of CPU time. This means an employee can check email while waiting for a spreadsheet to calculate, improving efficiency.
While the OS provides basic functionality, the business also needs utility software. Encryption software is essential to protect sensitive customer and

financial data — if a laptop is stolen, encrypted data cannot be accessed. Backup software is needed to protect against data loss from hardware failure or ransomware. These utilities provide security and protection that goes beyond what the OS offers by default.

Exam tip: Extended questions expect you to apply concepts to the scenario. Don't just define terms — explain why they matter for this specific business.

Worked examples summary

The 12 worked examples above cover the main question types you will encounter on systems software. Here is a summary of the key question patterns:

Example	Question type	Key skill
1	State reasons (2 marks)	Two distinct purposes of an OS
2	Explain interface (2 marks)	OS hides hardware complexity
3	Give examples (2 marks)	Match OS types correctly
4	Explain need (2 marks)	OS provides services for applications
5	Describe function (3 marks)	Memory: allocate, protect, virtual
6	Explain multitasking (3 marks)	Time slicing, context switching
7	Explain drivers (2 marks)	Translate OS commands to hardware
8	Explain virtual memory (3 marks)	Secondary storage as RAM extension
9	Explain encryption (3 marks)	Data unreadable without key
10	Explain defragmentation (3 marks)	Contiguous sectors, less head movement
11	Two utilities (4 marks)	Separate points for each utility
12	Extended analysis (6 marks)	Apply OS functions + utilities to scenario

Quick reference

Key terms – exam-ready definitions

Term	Definition (mark scheme style)
Operating system	Software that manages computer hardware and provides services for application programs; acts as an interface between user and hardware

Term	Definition (mark scheme style)
User interface	The means by which a user interacts with a computer (GUI, CLI, menu-driven, etc.)
Memory management	OS function that allocates RAM to programs, prevents programs accessing each other's memory, and manages virtual memory
Multitasking	Allows multiple programs to run simultaneously by rapidly switching CPU time between them
Peripheral management	OS function that controls input/output devices using device drivers
Device driver	Software that translates OS commands into instructions a specific hardware device understands
User management	OS function that manages accounts, passwords, permissions, and access levels
File management	OS function that organises files into folders, handles naming, and controls read/write operations
Virtual memory	Section of secondary storage used as extra RAM when physical RAM is full
Utility software	Programs designed to maintain, optimise, or secure a computer system
Encryption	Converting data into an unreadable format using a key; only those with the key can decrypt it
Defragmentation	Reorganising files on a hard disk so each file is stored in contiguous sectors
Data compression	Reducing file size; lossless preserves all data, lossy permanently removes some data
Backup	Copy of data that can be restored if the original is lost, corrupted, or deleted

OS functions at a glance

Function	What it does	Key exam points
User interface	Provides way to interact with computer	GUI vs CLI; WIMP
Memory management	Allocates and protects RAM	Allocation, protection, virtual memory

Function	What it does	Key exam points
Multitasking	Runs multiple programs "at once"	Time slicing, context switching
Peripheral management	Controls I/O devices	Device drivers, Plug and Play
User management	Manages accounts and permissions	Admin vs standard user, access levels
File management	Organises and controls files	Folders, naming, permissions

Utility software comparison

Utility	Purpose	Key point to remember
Encryption	Protect data from unauthorised access	Data is useless without the decryption key
Defragmentation	Improve HDD performance	Not needed for SSDs (no moving parts)
Compression	Reduce file sizes	Lossless = no data lost; Lossy = some data lost
Backup	Protect against data loss	Full vs incremental vs differential

Typical pitfalls

Pitfall 1 — Confusing the OS with application software.
Wrong: "Word is an operating system."
Right: Word is an **application**. Windows/macOS/Linux are operating systems. Applications run **on top of** the OS.

Pitfall 2 — Vague answers about OS purpose.
Wrong: "The OS runs the computer."
Right: The OS **manages hardware resources**, **provides a user interface**, and **provides services for applications**. Be specific.

Pitfall 3 — Confusing virtual memory with cloud storage.
Wrong: "Virtual memory is storage on the internet."
Right: Virtual memory is a section of **local secondary storage** (like the hard drive) used as extra RAM. It has nothing to do with the cloud.

Pitfall 4 — Saying multitasking means "running at the same time".

Wrong: "The CPU runs multiple programs at exactly the same time."
Right: A single-core CPU runs one instruction at a time. Multitasking **rapidly switches** between programs, **creating the illusion** of simultaneous execution.

Pitfall 5 — Forgetting that defragmentation is for HDDs only.
Wrong: "Defragmentation speeds up all storage devices."
Right: Defragmentation improves **HDD** performance only. SSDs don't need it because they have **no moving parts** and can access any location equally fast.

Pitfall 6 — Not explaining what drivers actually do.
Wrong: "Drivers help peripherals work."
Right: Drivers **translate OS commands** into instructions the specific hardware device understands. Different devices need different drivers.

Pitfall 7 — Confusing lossless and lossy compression.
Wrong: "Compression always loses quality."
Right: Lossless compression preserves all data (used for documents/programs). **Lossy** compression removes some data (used for images/audio where quality loss is acceptable).

Pitfall 8 — Not linking encryption to the key.
Wrong: "Encryption scrambles data so hackers can't read it."
Right: Encryption converts data to an unreadable format. **Only someone with the decryption key** can convert it back. Without the key, the data is useless.

Pitfall 9 — Confusing OS and utility software roles.
Wrong: "The OS always includes encryption and backup."
Right: Some OSes include basic versions of these tools, but **utility software** provides specialised or enhanced functions. The 2024 spec clarifies that utilities perform **additional tasks that may not be carried out by an OS**.

Pitfall 10 — Not applying answers to scenarios.
Wrong: "Memory management allocates RAM." (generic)
Right: "When the business runs multiple applications, the OS allocates RAM to each one so they can all function without crashing." (applied to scenario)

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